Binary search trees exercises

12-E01 Inorder and BST validation

Difficulty: Basic

Root 5 has left child 3 and right child 8; 3 has right child 6 and 8 has left child 7. Write the inorder traversal and decide whether this is a valid BST.

12-E02 Successor path

Difficulty: Advanced

In a BST, x=4 has no right child, parent 2, with 2 the left child of 6 (2.parent=6). Find 4's inorder successor and explain the upward stopping condition.

12-E03 Two-child deletion

Difficulty: Advanced

Delete 10 from a BST. Its right-subtree minimum is 12, and 12 has right child 13. Explain the replacements and transplants, checking parent pointers.

12-E04 Transplanting the successor's right child

Difficulty: Research

Insert 20,10,40,5,15,30,50,25,35,27 in order, then delete root20. Identify its successor, that node's old parent and right child, and all affected parent links. Explain why copying a key has different handle semantics.

12-E05 Uniform permutations versus uniform shapes

Difficulty: Advanced

List heights and shape multiplicities produced by the six insertion permutations of keys1,2,3. Compute expected edge height. What if each of the five shapes is equally likely instead?

12-E06 Average depth does not control maximum depth

Difficulty: Research

Start with a perfect balanced BST on N=2^h−1 keys. Append L=floor(sqrt(N)) increasing keys after the maximum-key leaf as a right chain. Estimate average node depth and height for the new n=N+L nodes. Why cannot an expected-height proof analyze only the depth of one uniform node?

Binary search trees exercises (continued)

> This file contains questions only. Complete your derivations before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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